Task 2: Data Types, Constraints & Table Design

Tools:

- MySQL Workbench
- Alternatives: PostgreSQL, SQLite, DBeaver

Hints / Mini Guide:

1. Redesign the students table using correct SQL data types like INT, VARCHAR, DATE.
2. Add constraints such as PRIMARY KEY, NOT NULL, UNIQUE.
3. Understand why constraints are essential for data integrity.
4. Try inserting invalid data to see constraint failures.
5. Modify the table using ALTER TABLE to add a new column.
6. Rename a column using SQL syntax.
7. Drop a column carefully and understand consequences.
8. Document why each constraint was chosen.

Deliverables:

- SQL script showing table creation with constraints and alterations

Final Outcome:

- Intern learns proper schema design and validation rules.

Interview Questions Related To Above Task:

- Difference between CHAR and VARCHAR?
- Why is PRIMARY KEY important?
- What is NOT NULL?
- Can a table have multiple unique constraints?
- What happens if we drop a column?

📌 **Task Submission Guidelines**

- ⏰ **Time Window:**

Youcan complete the task anytime between 10:00 AM to 10:00 PM on the given day. Submission link closes at 10:00 PM

- 🔍 **Self-Research Allowed:**

Youare free to explore, Google, or refer to tutorials to understand concepts and complete the task effectively.

- 🛠️ **Debug Yourself:**

Tryto resolve all errors by yourself. This helps you learn problem-solving and ensures you don't face the same issues in future tasks.

- 💸 **No Paid Tools:**

Ifthe task involves any paid software/tools, do not purchase anything. Just learn the process or find free alternatives.

- 📁 **GitHub Submission:**

Create a new GitHub repository for each task.
Add everything you used for the task — code, datasets, screenshots (if any), and a short README.md explaining what you did.

📤 **Submit Here:**

After completing the task, paste your GitHub repo link and submit it using the link below:

- 👉 [Submission Link]